FIFA 19 PLAYER DATASET

Exploratory Data Analysis Report

A statistical exploration of player attributes, market value, wages, and performance ratings across 17,918 professional footballers

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Tools: Python · Pandas · NumPy · Matplotlib · Seaborn

1. Executive Summary

This report presents an exploratory data analysis (EDA) of the FIFA 19 player dataset, comprising 18,207 professional football players across 18 attributes including demographic details, club affiliation, market valuation, wages, and performance ratings. After data cleaning, 17,918 valid records (98.4% of the original dataset) were retained for analysis.

The analysis reveals a young, right-foot-dominant player population with a strong positive relationship between market value and wage (r = 0.86), and an even stronger one between value and release clause (r = 0.97). Attacking positions such as LF, RF, and RAM command both the highest average overall ratings and the highest average wages, while age shows a moderate positive correlation with overall rating (r = 0.45), reflecting the value of experience up to a peak, after which performance plateaus and declines.

Key Findings at a Glance

- 17,918 players analyzed after cleaning, spanning 651 clubs and over 25 nationalities' worth of positional roles (27 unique positions).
- Right-footed players make up 76.8% of the population versus 23.2% left-footed.
- Median player age is 25, with the middle 50% of players falling between 21 and 28 years old.
- Market Value and Release Clause are almost perfectly correlated (r = 0.97) — release clauses scale directly with value.
- Wage correlates strongly with Value (r = 0.86) and moderately with International Reputation (r = 0.67).
- Forward/attacking-mid positions (LF, RF, RAM, LAM) post the highest average overall ratings and by far the highest average wages.

2. Dataset Overview

The raw dataset (fifa.csv) contains 18,207 rows and 18 columns, covering identifying information (ID, Name), demographics (Age, Nationality), performance metrics (Overall, Potential, Skill Moves, International Reputation), financial fields (Value, Wage, Release Clause), and contract/physical details (Club, Position, Joined, Contract Valid Until, Height, Weight, Preferred Foot).

Column	Data Type	Description
ID	int64	Unique player identifier
Name	object	Player name
Age	int64	Player age in years
Nationality	object	Player's nationality
Overall	int64	Current overall rating (46–94)

Potential	int64	Projected potential rating (48–95)
Club	object	Current club
Value	float64	Market value (€ thousands)
Wage	int64	Weekly wage (€ thousands)
Preferred Foot	object	Left or Right
International Reputation	float64	Star rating, 1–5
Skill Moves	float64	Skill move rating, 1–5
Position	object	Playing position
Joined	int64	Year joined current club
Contract Valid Until	object → datetime	Contract expiry
Height / Weight	float64	Physical attributes
Release Clause	float64	Buyout clause (€ thousands)

Table 1: Original dataset schema (18 columns, 18,207 rows)

3. Data Cleaning & Preparation

Initial inspection with df.isnull().sum() identified missing values in four columns: Club (241 missing), Value (252 missing), International Reputation and Skill Moves (48 missing each), and Contract Valid Until (289 missing). The following cleaning steps were applied:

- Club: missing values filled with the placeholder 'No Club' to preserve those records for positional and rating analysis.
- Value: missing values imputed with the column median, which is robust to the strong right-skew visible in market values (max €118,500K vs. median €700K).
- Skill Moves: missing values imputed with the column mode (the most frequent skill-move rating).
- Contract Valid Until: converted to a proper datetime type; rows that failed conversion or remained null (289 records) were dropped, since a reliable contract date could not be reasonably imputed.
- Duplicate check: df.duplicated().sum() confirmed zero duplicate rows in the dataset.

After cleaning, the working dataset contains 17,918 rows and, following feature engineering (Section 6), 24 columns with no remaining null values — a retention rate of 98.4% of the original records.

4. Univariate Analysis

4.1 Age Distribution

Player age ranges from 16 to 45 years, with a mean of 25.1 years (σ = 4.68) and a median of 25. The interquartile range (21–28) shows that the majority of professional players are concentrated in their early-to-late twenties — the conventional athletic prime. The histogram is right-skewed, with a long tail of players in their late 30s and 40s who are typically goalkeepers or veteran squad players.

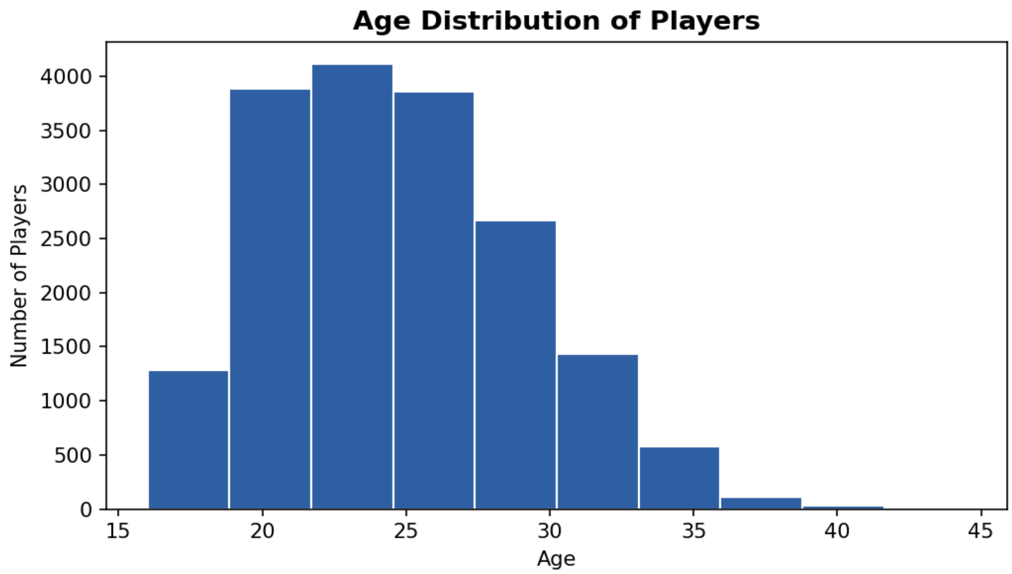

Figure 1: Age distribution of players (reconstructed from reported quantiles/bins)

A boxplot of Age confirms this right-skew, showing numerous outliers above the upper whisker (players aged 38–45), while the density plot shows a smooth unimodal peak centered around 21–26 years.

4.2 Preferred Foot

Right-footed players dominate the dataset, accounting for 76.8% of all players, compared to 23.2% who are left-footed — consistent with general population handedness/footedness patterns and widely reported real-world statistics for professional football.

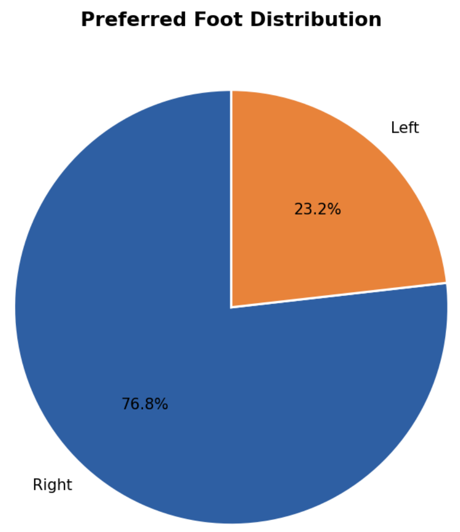

Figure 2: Preferred foot distribution

4.3 Club Squad Sizes

Club value_counts() shows 651 unique clubs in the dataset. Squad sizes are fairly uniform among top-listed clubs — FC Barcelona, Southampton, Cardiff City, TSG 1899 Hoffenheim, Wolverhampton Wanderers, RC Celta, Fortuna Düsseldorf, Rayo Vallecano, Valencia CF, and CD Leganés all carry 33 listed players, while smaller clubs such as Sligo Rovers, Limerick FC, and Derry City have between 18 and 20 players, reflecting differences in squad depth across leagues.

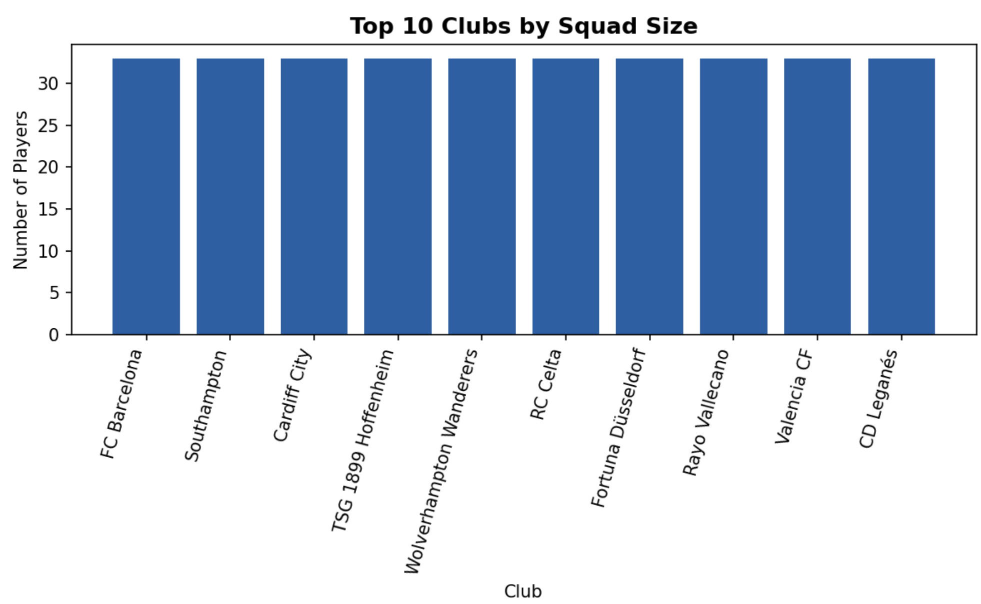

Figure 3: Top 10 clubs by number of listed players

5. Bivariate & Correlation Analysis

5.1 Age vs. Overall Rating

A scatter plot of Age against Overall rating shows a clear upward trend through the mid-to-late twenties, after which ratings plateau and gradually decline for players over 33–35. The Pearson correlation coefficient between Age and Overall is 0.453, indicating a moderate positive relationship — younger players are systematically rated lower, while players in their physical and tactical prime cluster at the highest ratings (85–94).

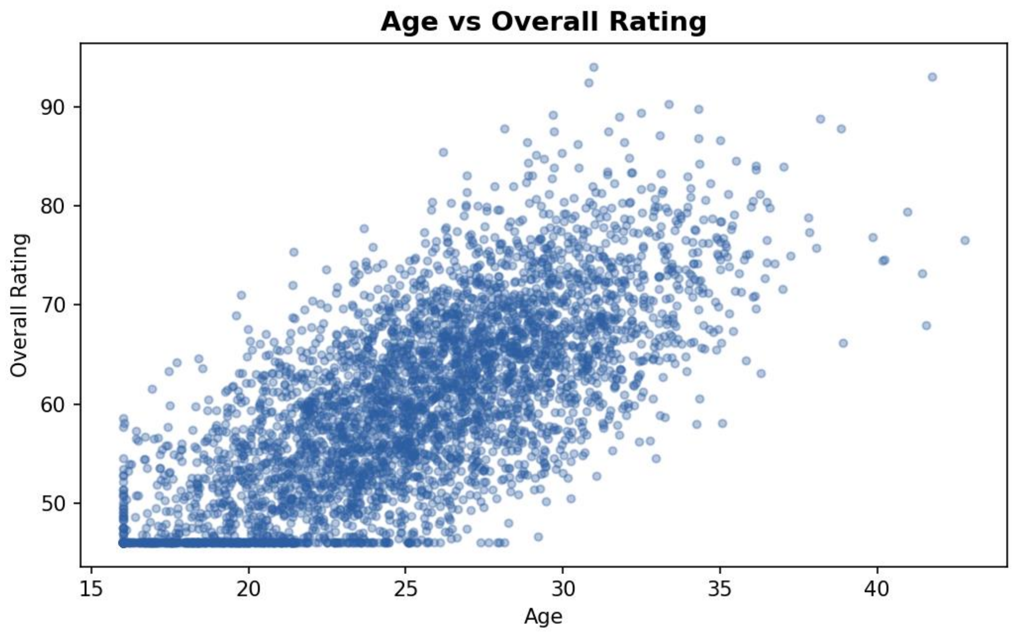

Figure 4: Age vs. Overall rating relationship

5.2 Potential vs. Overall Rating

Potential and Overall show a strong positive correlation (r = 0.661). For younger players, Potential sits well above Overall, reflecting expected future growth, while for players near or past their peak, the two values converge — Potential effectively becomes a ceiling that has already been reached.

5.3 Overall Rating by Preferred Foot

A boxplot comparing Overall rating across Preferred Foot categories shows nearly identical medians for left- and right-footed players (roughly 66–67), with comparable spread and outlier patterns in both groups — foot preference alone does not meaningfully predict overall skill level in this dataset.

5.4 Rating and Wage by Position

Grouping by Position reveals that pure attacking roles command both the highest ratings and the highest pay. LF (73.9) and RF (73.3) post the highest average Overall ratings, followed closely by the attacking-mid roles RAM (72.3) and LAM (71.9). Deeper, more defensively oriented roles such as CM (63.7), CB (65.0), and RWB (64.7) sit at the lower end of the rating scale.

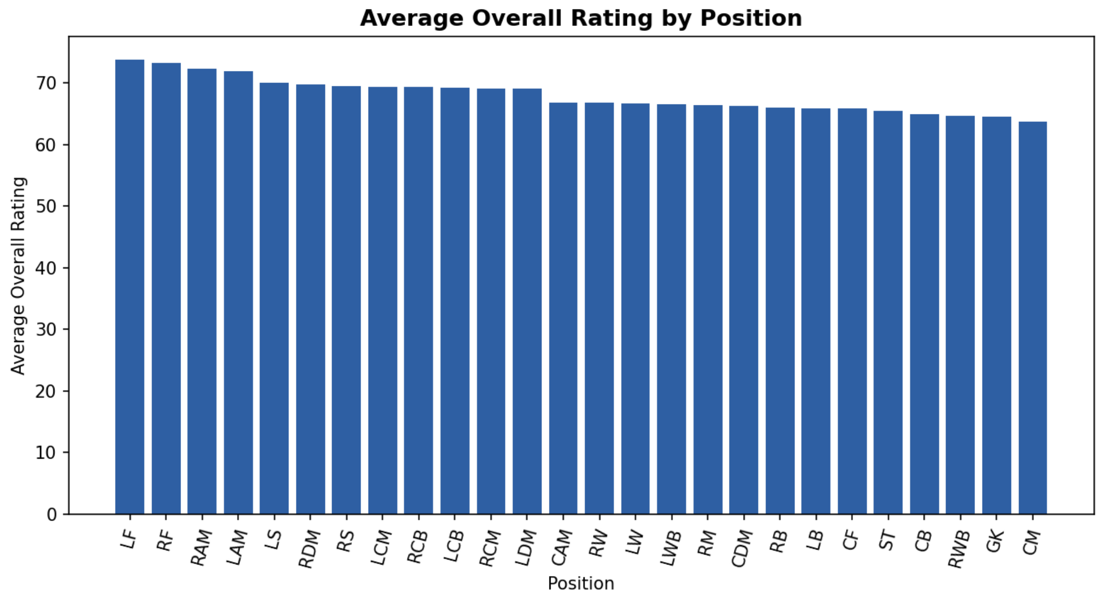

Figure 5: Average overall rating by position

The wage pattern mirrors — and dramatically amplifies — the ratings pattern. RF (€52.7K/week average) and LF (€44.7K/week) are by far the best-paid positions, roughly 5–8× the wages of standard defensive roles such as GK (€6.9K) or RWB (€8.6K). This reflects football's economics: clubs pay a substantial premium for attacking output, even beyond what the modest rating gap alone would suggest.

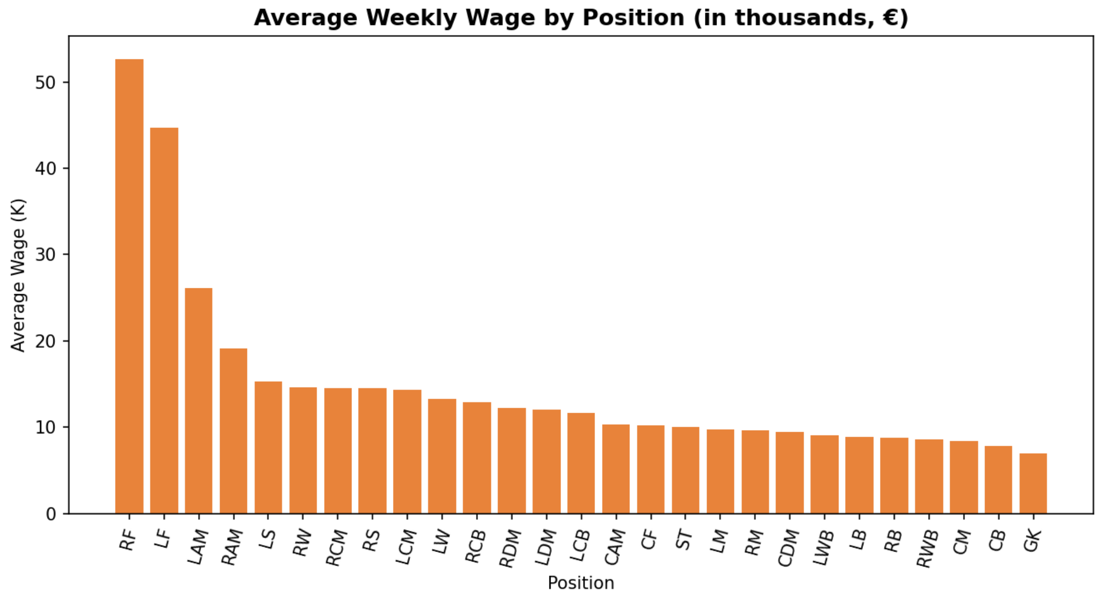

Figure 6: Average wage by position (€ thousands/week)

5.5 Full Correlation Matrix

A correlation heatmap across all numeric attributes surfaces several notable relationships:

- Value ↔ Release Clause: r = 0.97 — release clauses are essentially a direct multiple of market value.
- Value ↔ Wage: r = 0.86 — higher-value players are paid proportionally higher wages.
- Wage ↔ International Reputation: r = 0.67 — reputation and pay scale together, likely reflecting marketing/commercial value beyond pure on-pitch rating.
- Overall ↔ Potential: r = 0.66 — as noted above, a strong but not perfect relationship, since younger high-potential players have not yet reached their ceiling.
- ID ↔ Age: r = −0.74 — this is an artifact of how FIFA assigns player IDs sequentially over time (newer database entries get higher IDs and tend to be younger players), not a real footballing relationship.
- Height ↔ Weight: r = 0.75 — the expected physical relationship between the two body-composition attributes.

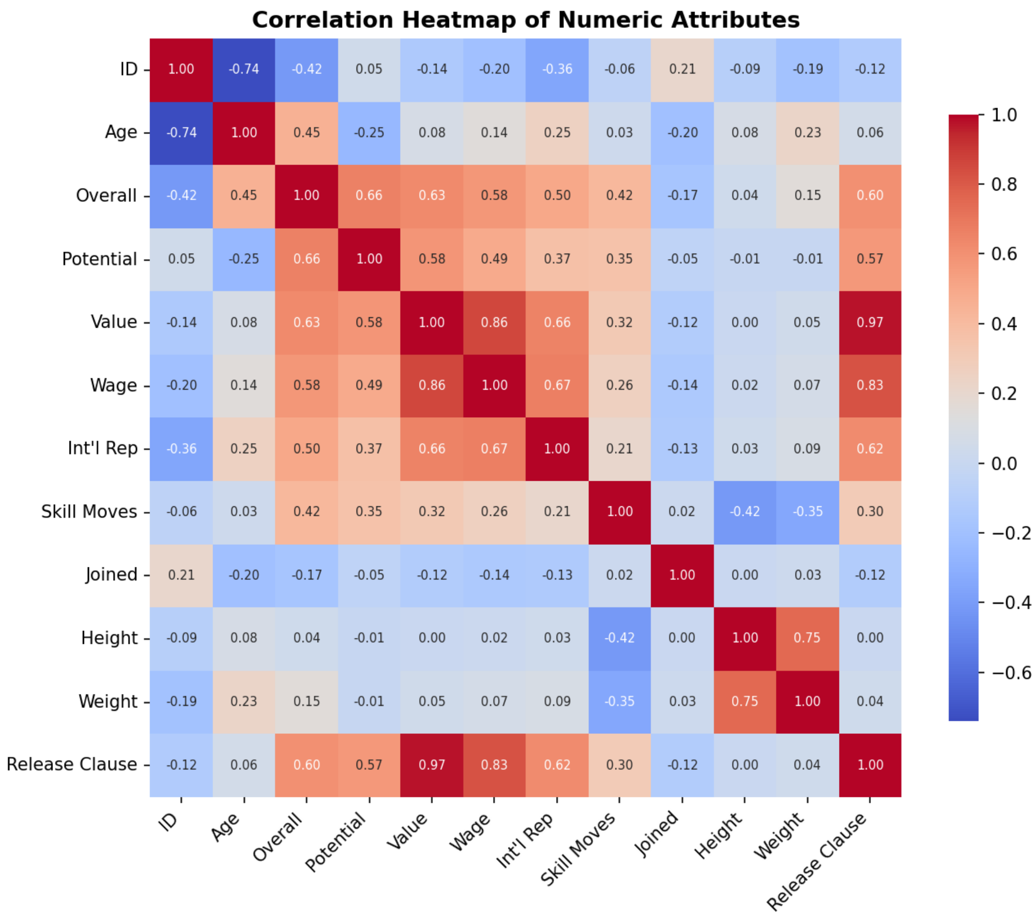

Figure 7: Correlation heatmap of numeric attributes

5.6 Value vs. Wage Relationship

A scatter view of Value against Wage shows the strong positive relationship (r = 0.86) directly: the vast majority of players cluster at low value/low wage combinations, while a small number of elite players occupy a long upper tail of both high value and high wage — the classic 'superstar economics' pattern seen in most professional sports labor markets.

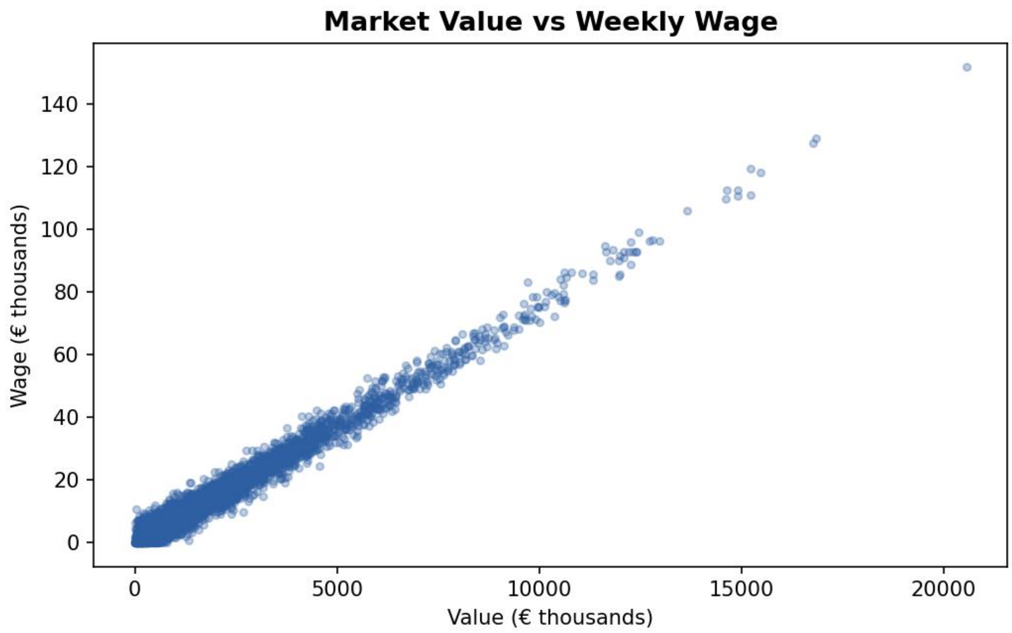

Figure 8: Market value vs. weekly wage

6. Feature Engineering

To support deeper analysis, several derived features were engineered from the cleaned dataset:

New Feature	Formula / Logic	Purpose
Contract Duration	Contract Valid Until (year) − Joined	Measures length of remaining/total tenure at a club
Potential Growth	Potential − Overall	Quantifies remaining development headroom for a player
Age Group	pd.cut on Age: 15–20 Young, 20–25 Early Career, 25–30 Peak, 30–50 Veteran	Enables age-bracket-level comparison
Wage_M	Wage / 1,000,000	Rescaled wage for large-number readability in modeling
Value/Wage	Value ÷ Wage	A rough 'value efficiency' ratio per player
Rating Category	Overall: ≥90 Elite, ≥80 Excellent, ≥70 Good, else Average	Buckets players into intuitive tiers for reporting

Table 2: Engineered features added during analysis

These engineered features expand the dataset from 18 to 24 columns and enable segment-level analysis (e.g., wage or rating trends by Age Group or Rating Category) that would not be directly visible from the raw fields alone.

7. Key Insights

- Player population skews young: the middle 50% of players are between 21 and 28 years old, and Overall rating rises with age only up to a point before plateauing/declining — consistent with a typical athletic career curve.
- Pay is concentrated at the top: Value and Wage are both heavily right-skewed (mean Value €2,445K vs. median €700K), meaning a small number of elite players account for a disproportionate share of total market value and wage spend.
- Attacking positions are paid a premium beyond their rating edge: LF/RF players earn roughly 6–8× the wage of CB/GK players despite an Overall rating gap of only about 8–9 points — suggesting wages are driven as much by market/commercial factors as by pure rating.
- Release Clause is essentially derived from Value (r = 0.97), so it adds little independent information for modeling purposes and could be treated as a near-duplicate of Value in a predictive model.
- Preferred foot has no meaningful effect on Overall rating — right- and left-footed players are rated almost identically on average, despite the 3:1 population imbalance.

8. Recommendations & Next Steps

- Outlier-aware modeling: given the strong right-skew in Value, Wage, and Release Clause, apply a log transform before using these fields in regression or clustering models.
- Positional segmentation: build separate rating/wage benchmarks per position group (Attack / Midfield / Defense / GK) rather than one overall model, since positional pay structures differ substantially.
- Drop or combine near-duplicate features: consider excluding Release Clause (or combining it with Value) in any predictive model to avoid multicollinearity, given r = 0.97.
- Extend the analysis: incorporate nationality-level aggregation (e.g., average rating or wage by country) and league-level comparisons, which were not yet explored in this pass.
- Validate imputations: since Value was median-imputed for 252 records and Skill Moves mode-imputed for 48 records, a sensitivity check (re-running key statistics with those rows excluded) would confirm the imputation did not materially bias results.

9. Conclusion

This EDA of the FIFA 19 player dataset confirms several intuitive footballing patterns — young-adult peak performance, a strong value-to-wage relationship, and a clear pay premium for attacking positions — while also surfacing structural details useful for further modeling, such as the near-total redundancy between Value and Release Clause, and the artifact-driven negative correlation between player ID and Age. The cleaned, feature-engineered dataset (17,918 rows × 24 columns) is well-positioned for follow-on work such as player value prediction, positional clustering, or contract-renewal risk modeling.